

Computational Mathematics and Statistics

Examination Solutions (E1PA101T)

Galgotias University | Mid-Term Nov 2025 | MCA Sem I

Section A: Matrix Theory

1. Define Upper Triangular Matrix. Give an example of order 4x4 and find its trace.

An Upper Triangular Matrix is a square matrix in which all the elements below the main diagonal are zero.

Example (4x4):

```
[ 2 5 7 1 ]  
[ 0 3 4 8 ]  
[ 0 0 6 2 ]  
[ 0 0 0 9 ]
```

Trace: The sum of the diagonal elements: $2 + 3 + 6 + 9 = 20$.

2. Find Inverse of Matrix $A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$ using Gauss-Jordan Method.

Step 1: Augment with Identity Matrix $[A|I]$.

Step 2: $R_2 \rightarrow R_2 - 2R_1$ gives $\begin{bmatrix} 1 & 3 & | & 1 & 0 \\ 0 & 1 & | & -2 & 1 \end{bmatrix}$.

Step 3: $R_1 \rightarrow R_1 - 3R_2$ gives $\begin{bmatrix} 1 & 0 & | & 7 & -3 \\ 0 & 1 & | & -2 & 1 \end{bmatrix}$.

Result: $A^{-1} = \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix}$.

Section B: Statistics & Probability

3. Calculate Mean and Median for data: {14, 16, 16, 16, 17, 16, 18}

Mean: $\text{Sum} (113) / \text{Count} (7) = 16.14$ approx.

Median: Arranging in ascending order: {14, 16, 16, 16, 16, 17, 18}. The middle value is 16.

4. Find Median from Grouped Data (Marks Table).

Total students (N) = 100. Median class is where cumulative frequency reaches $N/2$ (50). Based on frequencies [10, 20, 25, 20, 15, 10], the median class is 25-35.

$$\begin{aligned}\text{Median} &= L + \left[\frac{(N/2 - CF)}{f} \right] * h \\ &= 25 + \left[\frac{(50 - 30)}{25} \right] * 10 = 25 + 8 = 33.\end{aligned}$$

Section C: Numerical Methods

5. Solve by Gauss Elimination: $x-2y+z=0$; $2x+y-3z=5$; $4x-7y+z=-1$.

Step 1: Eliminate x from eq 2 and 3 using eq 1.

Step 2: Eliminate y from the new eq 3.

Step 3: Back substitution leads to: $x = 3$, $y = 2$, $z = 1$ (Values verified for consistency).

6. Algebraic and Geometric Multiplicity.

For matrix $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{bmatrix}$:

1. Find characteristic equation $\det(A - \lambda I) = 0$.
2. Algebraic Multiplicity (AM) is the number of times an eigenvalue repeats.
3. Geometric Multiplicity (GM) is the dimension of the eigenspace associated with that eigenvalue.